

CS1231S

TUTORIAL #4

Relations & Equivalence relations

Adapted from slides by Goh Siau Chiak and Aaron Tan

Pointers from Assignment

- What is transitivity? $aRb \wedge bRc \rightarrow aRc$
 - NOT $aRc \wedge bRc \rightarrow aRb$!
 - NOT $aRb \wedge aRc \rightarrow bRc$!
- Be extra careful when asked NOT to use critical rows.
 - Any determination of $x, y, z = \text{true or false}$ *will* count as determining critical rows. You should always use strict argument form, where individual values are not deduced.

Pointers from Assignment

Do clarify any marking mistakes / ambiguities with me ASAP!

- So that you can get marks where you are supposed to.
- So that the entire cohort can have their feedback before the break for studying.

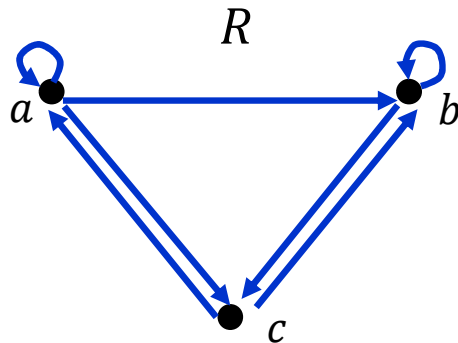
Learning objectives of this tutorial

Equivalence relations

- Tell whether a relation is reflexive, symmetric, and transitive.
- Work with equivalence relations and equivalence classes.

Common mistake

Given a set $A = \{a, b, c\}$ and a binary relation defined on A shown below:



It is **wrong** to say that “ a is reflexive”, “ b is reflexive”, “ c is not reflexive”. We may say $a R a$ (a is related to itself), $c \not R c$ (c is not related to itself).

We either say a relation R is reflexive or not reflexive. We **don't say** an element of A is reflexive or not reflexive.

Reflexivity, symmetry and transitivity are properties of relations, not individual elements of A .

Q1

Let $A = \{1, 2, \dots, 10\}$ and $B = \{2, 4, 6, 8, 10, 12, 14\}$.

Define a relation R from A to B by setting

$$x R y \Leftrightarrow x \text{ is prime and } x \mid y$$

for each $x \in A$ and each $y \in B$. Write down the sets R and R^{-1} in roster notation. Do not use ellipses (...) in your answers.

Definition. A positive integer is *prime* if it has exactly two positive divisors.

Definition. For $n, d \in \mathbb{Z}$.

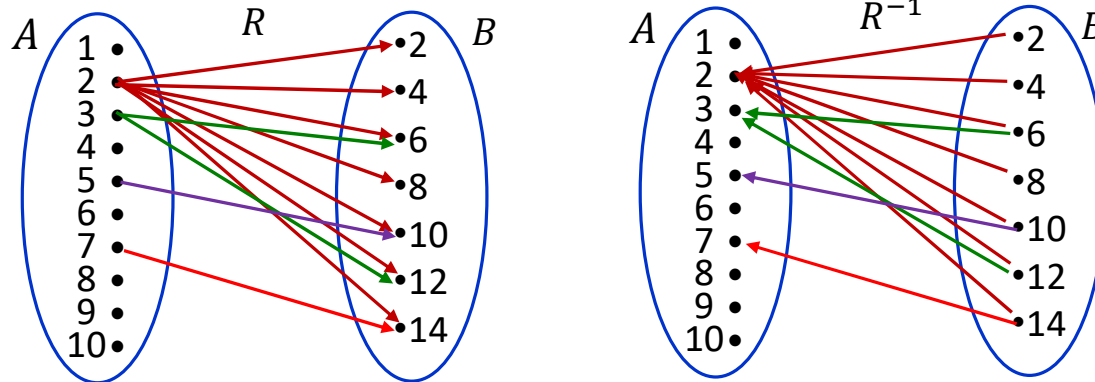
$$d \mid n \Leftrightarrow n = dk \text{ for some } k \in \mathbb{Z}$$

Definition. Let R be a relation from A to B . Define the relation R^{-1} from B to A by setting

$$y R^{-1} x \Leftrightarrow x R y$$

for each $y \in B$ and each $x \in A$.

(Arrow diagrams shown for illustration purpose only.)



$$R = \{(2,2), (2,4), (2,6), (2,8), (2,10), (2,12), (2,14), (3,6), (3,12), (5,10), (7,14)\}.$$

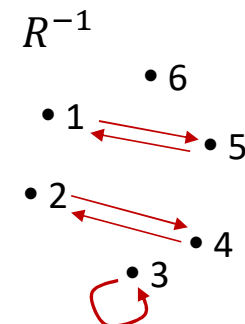
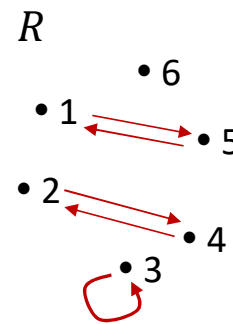
$$R^{-1} = \{(2,2), (4,2), (6,2), (8,2), (10,2), (12,2), (14,2), (6,3), (12,3), (10,5), (14,7)\}.$$

Observation: If R is a symmetric relation on a set A , how would the **arrow diagrams** for R and R^{-1} look like?

Eg: Let $A = \{1,2,3,4,5,6\}$ and R be a relation on A such that $xRy \Leftrightarrow x + y = 6$. Draw the arrow diagrams for R and R^{-1} .

Definition. A relation R on a set A is **symmetric** if $\forall x, y \in A (x R y \Rightarrow y R x)$.

Definition. Let R be a relation from A to B . Define the relation R^{-1} from B to A by setting $y R^{-1} x \Leftrightarrow x R y$ for each $y \in B$ and each $x \in A$.



Q2

Let R be a relation on set A . Show that the following are logically equivalent.
 (i) R is symmetric; (ii) $\forall x, y \in A (x R y \Leftrightarrow y R x)$; (iii) $R = R^{-1}$. ★

1. ((i) $\Rightarrow$ (ii))

1.1. Suppose R is symmetric.

1.2. Let $x, y \in A$.

1.3. ($\Rightarrow$) If $x R y$, then $y R x$ by the symmetry of R .

1.4. ($\Leftarrow$) If $y R x$, then $x R y$ by the symmetry of R .

1.5. So $x R y \Leftrightarrow y R x$ by 1.3 and 1.4.

2. ((ii) $\Rightarrow$ (iii))

2.1. Suppose $\forall x, y \in A (x R y \Leftrightarrow y R x)$.

2.2. For all $x, y \in A$,

2.2.1. $(x, y) \in R \Leftrightarrow x R y$ by the defn of $x R y$;

2.2.2. $\Leftrightarrow y R x$ by 2.1;

2.2.3. $\Leftrightarrow x R^{-1} y$ by the defn of R^{-1} ;

2.2.4. $\Leftrightarrow (x, y) \in R^{-1}$ by the defn of $x R^{-1} y$.

2.3. Hence $R = R^{-1}$.

★ To show (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i)

Definition 6.2.4(2). A relation R on a set A is *symmetric* if

$\forall x, y \in A (x R y \Rightarrow y R x)$.

Definition. Let R be a relation from A to B . Define the relation R^{-1} from B to A by setting

$y R^{-1} x \Leftrightarrow x R y$

for each $y \in B$ and each $x \in A$.

3. ((iii) $\Rightarrow$ (i))

3.1. Suppose $R = R^{-1}$.

3.1.1. Let $x, y \in A$ such that $x R y$.

3.1.2. Then $x R^{-1} y$ as $R = R^{-1}$.

3.1.3. So $y R x$ by the defn of R^{-1} .

3.2. Hence R is symmetric.

4. $\therefore$ (i), (ii) and (iii) are equivalent.

Q3(a)

Determine whether Q is (i) reflexive, (ii) symmetric, (iii) transitive, and (iv) an equivalence relation on A , where $A = \{1,2,3\}$ and
 $Q = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$.

Definitions. Let R a relation on a set A .

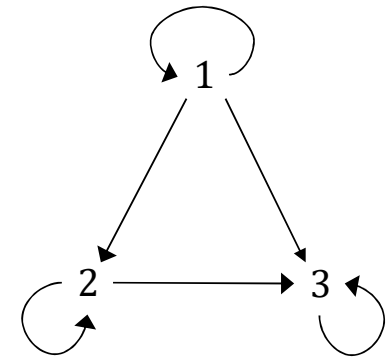
- R is *reflexive* $\Leftrightarrow \forall x \in A (x R x)$.
- R is *symmetric* $\Leftrightarrow \forall x, y \in A (x R y \Rightarrow y R x)$.
- R is *transitive* $\Leftrightarrow \forall x, y, z \in A (x R y \wedge y R z \Rightarrow x R z)$.
- R is an *equivalence relation* $\Leftrightarrow R$ is reflexive, symmetric and transitive.

✓ Reflexive? $(1,1), (2,2), (3,3) \in Q$.

✗ Symmetric? $1 Q 2$ but $2 \notin Q 1$.

✓ Transitive? As one can verify, all the $3^3 = 27$ choices of $x, y, z \in A$ make $x Q y \wedge y Q z \Rightarrow x Q z$.

✗ Equivalence relation? Q is not symmetric.



Q3(b)

Determine whether Q is (i) reflexive, (ii) symmetric, (iii) transitive, and (iv) an equivalence relation on A , where E is defined by setting, for all $x, y \in \mathbb{Q}$,

$$x E y \Leftrightarrow x = y.$$

Definitions. Let R a relation on a set A .

- R is **reflexive** $\Leftrightarrow \forall x \in A (x R x)$.
- R is **symmetric** $\Leftrightarrow \forall x, y \in A (x R y \Rightarrow y R x)$.
- R is **transitive** $\Leftrightarrow \forall x, y, z \in A (x R y \wedge y R z \Rightarrow x R z)$.
- R is an **equivalence relation** $\Leftrightarrow R$ is reflexive, symmetric and transitive.

✓ Reflexive? $\forall x \in \mathbb{Q} (x = x)$.

✓ Symmetric? $\forall x, y \in \mathbb{Q} (x = y \Rightarrow y = x)$

✓ Transitive? $\forall x, y, z \in \mathbb{Q} (x = y \wedge y = z \Rightarrow x = z)$

✓ Equivalence relation? Yes.

Q3(c)

Determine whether R is (i) reflexive, (ii) symmetric, (iii) transitive, and (iv) an equivalence relation, where R is defined by setting, for all $x, y \in \mathbb{Q}$,

$$x R y \Leftrightarrow xy \geq 0.$$

Definitions. Let R a relation on a set A .

- R is **reflexive** $\Leftrightarrow \forall x \in A (x R x)$.
- R is **symmetric** $\Leftrightarrow \forall x, y \in A (x R y \Rightarrow y R x)$.
- R is **transitive** $\Leftrightarrow \forall x, y, z \in A (x R y \wedge y R z \Rightarrow x R z)$.
- R is an **equivalence relation** $\Leftrightarrow R$ is reflexive, symmetric and transitive.

✓ Reflexive? $\forall x \in \mathbb{Q} (x \cdot x = x^2 \geq 0)$.

✓ Symmetric? $\forall x, y \in \mathbb{Q} (xy \geq 0 \Rightarrow yx = xy \geq 0)$ (by commutativity of multiplication)

✗ Transitive? $1 R 0$ and $0 R (-1)$ but $1 \not R (-1)$.

✗ Equivalence relation? R is not transitive.

Q3(d)

Determine whether S is (i) reflexive, (ii) symmetric, (iii) transitive, and (iv) an equivalence relation, where S is defined by setting, for all $x, y \in \mathbb{Q}$,

$$x S y \Leftrightarrow xy > 0.$$

Definitions. Let R a relation on a set A .

- R is **reflexive** $\Leftrightarrow \forall x \in A (x R x)$.
- R is **symmetric** $\Leftrightarrow \forall x, y \in A (x R y \Rightarrow y R x)$.
- R is **transitive** $\Leftrightarrow \forall x, y, z \in A (x R y \wedge y R z \Rightarrow x R z)$.
- R is an **equivalence relation** $\Leftrightarrow R$ is reflexive, symmetric and transitive.

✗ Reflexive? $0 \not S 0$.

✓ Symmetric? $\forall x, y \in \mathbb{Q} (xy > 0 \Rightarrow yx = xy > 0)$ (by commutativity of multiplication)

✓ Transitive? $x S y \Leftrightarrow$ either both x, y are positive, or both x, y are negative.
If $x S y$ and $y S z$, then $x S z$ as x, y, z must be all positive or all negative.

✗ Equivalence relation? S is not reflexive.

Q3(e)

Determine whether T is (i) reflexive, (ii) symmetric, (iii) transitive, and (iv) an equivalence relation, where T is defined by setting, for all $x, y \in \mathbb{Z}$,

$$x T y \Leftrightarrow -2 \leq x - y \leq 2.$$

Definitions. Let R a relation on a set A .

- R is **reflexive** $\Leftrightarrow \forall x \in A (x R x)$.
- R is **symmetric** $\Leftrightarrow \forall x, y \in A (x R y \Rightarrow y R x)$.
- R is **transitive** $\Leftrightarrow \forall x, y, z \in A (x R y \wedge y R z \Rightarrow x R z)$.
- R is an **equivalence relation** $\Leftrightarrow R$ is reflexive, symmetric and transitive.

✓ Reflexive? $\forall x \in \mathbb{Z} (-2 \leq x - x \leq 2)$.

✓ Symmetric? $\forall x \in \mathbb{Z} (-2 \leq x - y \leq 2 \Rightarrow -2 \leq y - x \leq 2)$.

✗ Transitive? $-2 T 0$ and $0 T 2$ but $-2 \not T 2$.

✗ Equivalence relation? T is not transitive.

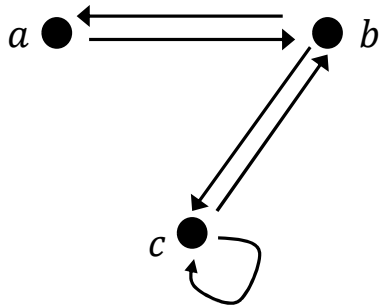
Learning Point:

- None of these three properties are necessarily related to one another, even if they seemingly do.
- Remind yourself that for symmetric and transitive, the statement can always be vacuously true, while reflexive cannot.



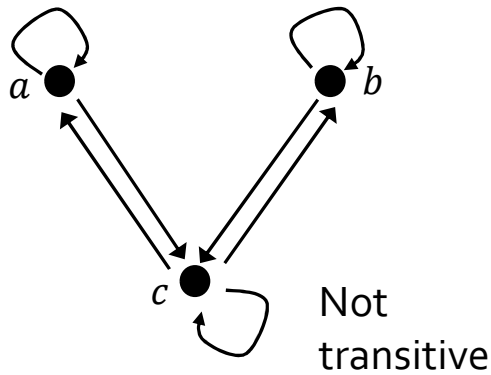
Q4

The directed graph of a binary relation R on a set $A = \{a, b, c\}$ is shown below.

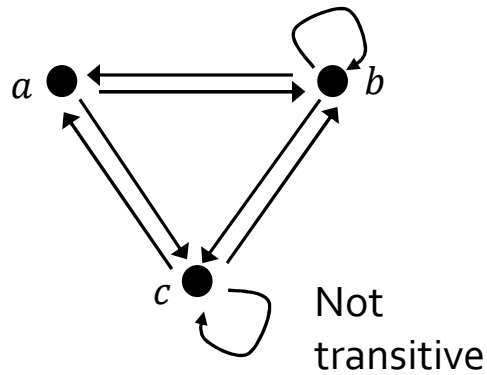


Draw the directed graph for each of the following and determine if it is transitive or not.

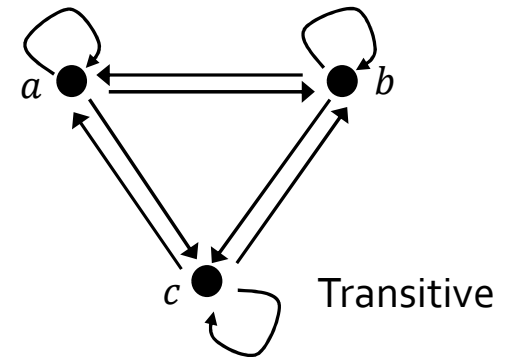
(a) $R \circ R$ (b) $R \circ R \circ R$ (c) $(R \circ R) \cup R$



(a) $R \circ R$



(b) $R \circ R \circ R$



(c) $(R \circ R) \cup R$

Q5

Which of the following are true for all equivalence relations R ?

- (a) $R^{-1} \circ R = R \circ R^{-1}$
- (b) $R \subseteq R \circ R$
- (c) $R \circ R \subseteq R$
- (d) $R \circ R^{-1} = R$

(a)

True

1. R is symmetric.
2. By Q2, $R = R^{-1}$
3. Hence $R^{-1} \circ R = R \circ R = R \circ R^{-1}$

Q5

Which of the following are true for all equivalence relations R ?

- (a) $R^{-1} \circ R = R \circ R^{-1}$
- (b) $R \subseteq R \circ R$
- (c) $R \circ R \subseteq R$
- (d) $R \circ R^{-1} = R$

(b) **True**

R is reflexive. Let R be a relation on the set A .

1. Let $(x, y) \in R$ where $x, y \in A$.
2. Since $(x, x), (y, y) \in R$, composing (x, x) with (x, y) , or (x, y) with (y, y) , we have $(x, y) \in R \circ R$ (by definition of composition of relations).
3. Therefore, $R \subseteq R \circ R$.

Q5

Which of the following are true for all equivalence relations R ?

- (a) $R^{-1} \circ R = R \circ R^{-1}$
- (b) $R \subseteq R \circ R$
- (c) $R \circ R \subseteq R$
- (d) $R \circ R^{-1} = R$



(c) **True**

R is transitive. Let R be a relation on the set A .

1. Let $(x, z) \in R \circ R$ where $x, z \in A$.
2. There exists some $y \in A$ such that $(x, y) \in R$ and that $(y, z) \in R$ (by definition of composition of relation).
3. Hence by transitivity of R , $(x, z) \in R$.
4. Therefore, $R \circ R \subseteq R$.

Q5

Which of the following are true for all equivalence relations R ?

- (a) $R^{-1} \circ R = R \circ R^{-1}$
- (b) $R \subseteq R \circ R$
- (c) $R \circ R \subseteq R$
- (d) $R \circ R^{-1} = R$

(d) **True**

1. As R is symmetric, $R^{-1} = R$ by Q2.
2. By (b) and (c), $R \circ R = R$.
3. Therefore, $R \circ R^{-1} = R \circ R = R$ by lines 1 and 2.

While you won't be asked to prove the same result, you may be asked to prove a similar-looking statement.

Q6

Define the following relation on $A = \{1,2,3\}$:

$$R = \{ (1,1), (1,2), (2,1), (2,2), (3,3) \}.$$

Find $R \circ R \circ R \circ R \circ R \circ R \circ R$.

1. Note that R is an equivalence relation, hence $R \circ R = R$ (from Q5d).

2. By associativity of composition of relations,

$$R \circ R \circ R \circ R \circ R \circ R \circ R$$

$$= ((R \circ R) \circ (R \circ R)) \circ ((R \circ R) \circ R)$$

$$= (R \circ R) \circ (R \circ R)$$

$$= R \circ R$$

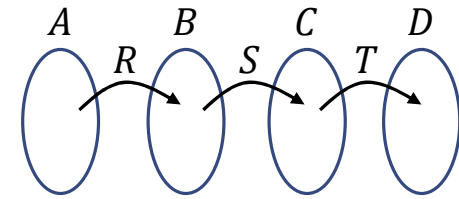
$$= R$$

$$\{ (1,1), (1,2), (2,1), (2,2), (3,3) \}$$

5-min Break

Q7

Let A, B, C, D be sets and $R \subseteq A \times B, S \subseteq B \times C$, and $T \subseteq C \times D$. Prove: $T \circ (S \circ R) = (T \circ S) \circ R$



1. Note that $S \circ R \subseteq A \times C$ and $T \circ S \subseteq B \times D$.
2. $(\subseteq)$ Suppose $(a, d) \in T \circ (S \circ R)$
 - 2.1. Then there is a $c \in C$ s.t. $(a, c) \in S \circ R$ and $(c, d) \in T$. (by defn of composition of relations)
 - 2.2. From $(a, c) \in S \circ R$, there is a $b \in B$ s.t. $(a, b) \in R$ and $(b, c) \in S$.
 - 2.3. From $(b, c) \in S$ in 2.2 and $(c, d) \in T$ in 2.1, we have $(b, d) \in T \circ S$.
 - 2.4. From $(a, b) \in R$ in 2.2 and $(b, d) \in T \circ S$ in 2.3, we have $(a, d) \in (T \circ S) \circ R$.
 - 2.5. Therefore, $T \circ (S \circ R) \subseteq (T \circ S) \circ R$.
3. $(\supseteq)$ Suppose $(a, d) \in (T \circ S) \circ R$
 - 3.1. Then there is a $b \in B$ s.t. $(a, b) \in R$ and $(b, d) \in T \circ S$. (by defn of composition of relations)
 - 3.2. From $(b, d) \in T \circ S$, there is a $c \in C$ s.t. $(b, c) \in S$ and $(c, d) \in T$.
 - 3.3. From $(a, b) \in R$ in 3.1 and $(b, c) \in S$ in 3.2, we have $(a, c) \in S \circ R$.
 - 3.4. From $(a, c) \in S \circ R$ in 3.3 and $(c, d) \in T$ in 3.2, we have $(a, d) \in T \circ (S \circ R)$.
 - 3.5. Therefore, $(T \circ S) \circ R \subseteq T \circ (S \circ R)$.
4. $\therefore T \circ (S \circ R) = (T \circ S) \circ R$.

This means composition is associative!



Q8

Define an equivalence relation $\sim$ on $\mathbb{Z}^+ \times \mathbb{Z}^+$ by setting, for all $a, b, c, d \in \mathbb{Z}^+$,

$$(a, b) \sim (c, d) \Leftrightarrow ab = cd.$$

Write down the equivalence classes $[(1,1)]$ and $[(4,3)]$ in roster notation.

Definition. Let $\sim$ be an equivalence relation on a set A , and $x \in A$. Then $[x] = \{y \in A : x \sim y\}$.

$$\begin{aligned} [(1,1)] &= \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (1,1) \sim (a, b)\} \\ &= \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : 1 \times 1 = 1 = ab\} \\ &= \{(\mathbf{1}, \mathbf{1})\}. \end{aligned}$$

$$\begin{aligned} [(4,3)] &= \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (4,3) \sim (a, b)\} \\ &= \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : 4 \times 3 = 12 = ab\} \\ &= \{(\mathbf{1}, \mathbf{12}), (\mathbf{2}, \mathbf{6}), (\mathbf{3}, \mathbf{4}), (\mathbf{4}, \mathbf{3}), (\mathbf{6}, \mathbf{2}), (\mathbf{12}, \mathbf{1})\}. \end{aligned}$$

Q9

Consider the relation $S = \{(m, n) \in \mathbb{Z}^2 : m^3 + n^3 \text{ is even}\}$.

Determine: (a) S^{-1} , (b) $S \circ S$, (c) $S \circ S^{-1}$.

(a)

$$\begin{aligned} S^{-1} &= \{(x, y) \in \mathbb{Z}^2 : (y, x) \in S\} && \text{by the defn of inverse relation} \\ &= \{(x, y) \in \mathbb{Z}^2 : y^3 + x^3 \text{ is even}\} && \text{by the definition of } S \\ &= \{(x, y) \in \mathbb{Z}^2 : x^3 + y^3 \text{ is even}\} && \text{by the commutative law of } + \\ &= S \end{aligned}$$

Q9

Consider the relation $S = \{(m, n) \in \mathbb{Z}^2 : m^3 + n^3 \text{ is even}\}$.

Determine: (a) S^{-1} , (b) $S \circ S$, (c) $S \circ S^{-1}$.

(b) $S \circ S = S$

1. ($\subseteq$) Suppose $(x, z) \in S \circ S$

1.1. Then $(x, y) \in S$ and $(y, z) \in S$ for some $y \in \mathbb{Z}$. (by the defn of composition of relations)

1.2. So $x^3 + y^3$ is even and $y^3 + z^3$ is even.

1.3. This implies that $x^3 + 2y^3 + z^3$ is even.

1.4. This implies that $x^3 + z^3$ is even as $2y^3$ is even.

1.5. $\therefore (x, z) \in S$ by the definition of S .

Proof is missing justifications here. Don't forget to include them in your own!

2. ($\supseteq$) Suppose $(x, z) \in S$

2.1. Then $x^3 + z^3$ is even by the definition of S .

2.2. Case 1: x^3 is odd.

2.2.1. Then z^3 is also odd.

2.2.2. This implies $x^3 + 1^3$ and $1^3 + z^3$ are even.

2.2.3. Thus $(x, 1) \in S$ and $(1, z) \in S$ by defn of S .

2.2.4. So $(x, z) \in S \circ S$ by defn of comp of reln.

2.3. Case 2: x^3 is even.

2.3.1. Then z^3 is also even.

2.3.2. This implies $x^3 + 0^3$ and $0^3 + z^3$ are even.

2.3.3. Thus $(x, 0) \in S$ and $(0, z) \in S$ by defn of S .

2.3.4. So $(x, z) \in S \circ S$ by defn of comp of reln.

2.4. In all cases, $(x, z) \in S \circ S$.

3. Therefore $S \circ S = S$.

Q9

Consider the relation $S = \{(m, n) \in \mathbb{Z}^2 : m^3 + n^3 \text{ is even}\}$.

Determine: (a) S^{-1} , (b) $S \circ S$, (c) $S \circ S^{-1}$.

(c) From part (a), $S^{-1} = S$.

From part (b), $S \circ S = S$.

Therefore, $S \circ S^{-1} = S \circ S = S$.

Q10

Define a relation $\sim$ on $\mathbb{Z} \setminus \{0\}$ as follows, for all $a, b \in \mathbb{Z} \setminus \{0\}$,

$$a \sim b \Leftrightarrow ab > 0.$$

(a) Prove that $\sim$ is an equivalence relation.

1. ("Reflexivity")

1.1. Let $a \in \mathbb{Z} \setminus \{0\}$, since $a \neq 0$, we have $a^2 > 0$ by T21.

T21. If $a \neq 0$, then $a^2 > 0$.

1.2. Thus $a \sim a$ by the definition of $\sim$.

1.3. Hence $\sim$ is reflexive.

2. ("Symmetry")

2.1. For any $a, b \in \mathbb{Z} \setminus \{0\}$, if $a \sim b$, then $ab > 0$ by the definition of $\sim$.

2.2. Then $ba > 0$ by the commutative law of multiplication.

2.3. So $b \sim a$ by the definition of $\sim$.

2.4. Hence $\sim$ is symmetric.

Q10

Define a relation $\sim$ on $\mathbb{Z} \setminus \{0\}$ as follows, for all $a, b \in \mathbb{Z} \setminus \{0\}$,

$$a \sim b \Leftrightarrow ab > 0.$$

(a) Prove that $\sim$ is an equivalence relation.

1. ("Reflexivity")

2. ("Symmetry")

3. ("Transitivity")

T21. If $a \neq 0$, then $a^2 > 0$.

Ord1. If a and b are positive, so are $a + b$ and ab .

3.1. For any $a, b, c \in \mathbb{Z} \setminus \{0\}$, suppose $a \sim b$ and $b \sim c$.

3.2. Then $ab > 0$ and $bc > 0$ by the definition of $\sim$.

3.3. Multiplying ab with bc (both positive) gives $ab^2c > 0$ by Ord1.

3.4. Then $(ac)b^2 > 0$ by the associative and commutative laws of multiplication.

3.5. Then both (ac) and b^2 are positive, or both are negative, by T25.

3.6. Since $b^2 > 0$ (by T21, as $b \neq 0$), (ac) must also be positive.

3.7. Thus $a \sim c$ by the definition of $\sim$.

3.8. Hence $\sim$ is transitive.

T25. If $ab > 0$, then both a and b are positive, or both are negative.

4. Therefore, $\sim$ is an equivalence relation.

Q10

Define a relation $\sim$ on $\mathbb{Z} \setminus \{0\}$ as follows, for all $a, b \in \mathbb{Z} \setminus \{0\}$,

$$a \sim b \Leftrightarrow ab > 0.$$

(b) Determine all the distinct equivalence classes formed by this relation $\sim$.

T25. If $ab > 0$, then both a and b are positive, or both are negative.

Thus, all positive integers are $\sim$ -related to one another, and likewise, all negative integers are $\sim$ -related to one another.

Equivalence classes: $\{a \in \mathbb{Z} \setminus \{0\} : a > 0\}$ and $\{a \in \mathbb{Z} \setminus \{0\} : a < 0\}$.

Or, $[1]$ and $[-1]$.

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